Coherence time in biological oscillator assemblies bounds the rate of state registration

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Abstract

Cognitive and biological processing depends on coherent oscillations across distributed assemblies. Yet the physical principles determining how fast these assemblies can synchronize—and thus how fast neural processes associated with cognition can proceed—remain incompletely formalized. We derive a quantitative framework showing that coherence time in coupled oscillator networks scales exponentially with coordination depth. For M semi-independent modules requiring phase alignment within tolerance ε at coherence r, the minimum interval between irreversible state registrations ("commits") is dominated by rare-event first-passage time into low-measure alignment sets on the configuration manifold. This produces a fundamental speed-flexibility trade-off: increasing coordination depth expands combinatorial flexibility but slows commits exponentially; tighter coherence accelerates synchronization but restricts dynamics to low-dimensional attractors. We situate this coordination limit within a hierarchy of physical bounds—quantum speed limits, Landauer-Margolus-Levitin thermodynamic constraints, and energy-time uncertainty relations—showing that in biological neural systems, these quantum and thermodynamic bounds are typically non-binding; the dominant constraint is the time required for distributed oscillatory assemblies to achieve sufficient phase coherence. While neural oscillators provide the most empirically accessible measurements, the framework applies to any biological system whose function requires coordinated phase relations and sparse commitment events, from chromosomal segregation to circadian networks. We provide parameter estimation protocols and testable predictions including alpha entrainment effects on temporal acuity and dual-loop dissociations under arousal.

Keywords: biological oscillations, coherence time, coordination depth, temporal resolution, quantum speed limits, Landauer bound, coupled oscillators

1 Introduction

1.1 Processing speed in biological oscillator assemblies

A fundamental insight from systems neuroscience is that cognition emerges from coherent oscillations across neural assemblies, not merely from individual spike rates. Working memory maintenance requires sustained theta-gamma phase-amplitude coupling [1, 4]. Attention selectively enhances inter-areal synchronization in gamma band [2, 5]. Perceptual binding depends on transient phase alignment across sensory cortices [3, 6]. Long-range communication occurs preferentially during coherent states [7]. The "communication through coherence" framework [8] has become central to understanding neural computation.

Yet despite extensive empirical characterization, the physical principles governing *how fast* distributed assemblies can achieve coherence—and thus how quickly the neural processes underlying cognitive operations can proceed—remain incompletely formalized. Why does perceptual binding require 30–50 ms rather than 3 ms or 300 ms? Why do larger assemblies integrating more information process more slowly? What determines the relationship between oscillation frequency and temporal acuity?

1.2 Coherence time as a physical bound on processing rate

Previous work established that living intelligence is an emergent property of high-dimensional, self-sustaining dynamical systems that maintain coherence against decoherence [9]. The present manuscript addresses the complementary question: given such a coherent high-dimensional system, what sets the fastest timescale on which it can register change and update its state?

We formalize this as the minimum interval between irreversible "commit" events—transitions in which distributed dynamics become readable to downstream mechanisms. This commit interval is bounded by a coordination term (coherence time as a first-passage problem on a product manifold) together with quantum, noise, and power limits.

Definition (Ontological dimensionality). Throughout this work, "dimensionality" refers to a physical property of the system, not a mathematical measure. Ontological dimensionality is a quantitative measure of effective connectivity: how many independent directions of coordinated variation are accessible to a system given its interaction structure. High-dimensional states correspond to weakly constrained dynamics in which many components vary semi-independently; low-dimensional states correspond to strongly coupled dynamics restricted to a small number of collective modes. Dimensionality in this sense is determined not by component count, but by the rank of the system's interaction structure. Different measurement schemes (participation ratios, manifold dimension, information-geometric metrics) provide projections of this underlying physical property, but do not constitute it [10].

This ontological dimensionality is state-dependent: the same physical system may occupy high- or low-dimensional regimes depending on coupling strength and task demands. Coherence time depends on ontological dimensionality—specifically, how many semi-independent constraints must be simultaneously satisfied for a commit to occur—not on any particular estimator.

1.3 Structure of this work

We first derive the coherence time formula from principles of phase synchronization in coupled oscillator networks (§2.1), then embed it in a unified temporal resolution bound combining quantum, noise, and power constraints (§2.2). Section 2.3 situates our framework explicitly within the quantum and thermodynamic limits of computation, addressing the Landauer bound, Margolus-Levitin quantum speed limit, and energy-time uncertainty relation. Section 3 applies the framework to neural systems, demonstrating quantitative agreement with perceptual binding windows, tachypsychia dissociations, and metabolic scaling. Section 4 discusses the speed-flexibility trade-off, delay embedding as a preservation mechanism, generalization beyond neural systems, and testable predictions.

2 Theory and Methods

2.1 The unified temporal resolution bound

We model biological temporal processing as a sequence of *commits*—thermodynamically irreversible events that register high-dimensional internal state as low-dimensional output. A commit requires: (1) dimensional collapse from distributed state (D_{eff} dimensions) to registered outcome (few dimensions), (2) dissipation of at least $k_{\text{B}}T \ln 2$ per bit (Landauer bound), and (3) creation of a persistent measurement influencing future dynamics.

The minimum time between commits is bounded by four physical constraints:

$$\boxed{\tau_{\text{eff}} \;=\; \max\left[\tau_{\text{QSL}},\; \tau_{\text{SNR}},\; \tau_{\text{coh}},\; \tau_{\text{power}}\right]} \tag{1}$$

where the **max** operation reflects that the slowest mechanism dominates. The four terms are:

Quantum speed limits ($\tau_{\rm QSL}$). The Mandelstam-Tamm and Margolus-Levitin bounds [13, 14] establish

$$\tau_{\rm QSL} = \max\left[\frac{\hbar}{2\Delta E}, \frac{\pi\hbar}{2\langle E\rangle}\right]$$

For biological temperatures ($\Delta E \sim k_{\rm B}T$), $\tau_{\rm QSL} \sim 10^{-13}$ s.

Signal-to-noise limit ($\tau_{\rm SNR}$). Matched-filter detection in additive white Gaussian noise (two-sided power spectral density N_0, units: power/Hz) over bandwidth B achieves $\mathrm{SNR}(T) = 2P_sT/N_0$ where P_s is signal power and T is integration time [19]. Reaching detection threshold $\rho_{\rm min} \sim 5$–10 requires

$$\tau_{\rm SNR} = \frac{\rho_{\rm min} N_0}{2P_s}$$

Power limit ($\tau_{\rm power}$). Dimensional collapse by $\Delta\mathcal{D}$ at dimensional chemical potential $\lambda_{\mathcal{D}} \sim k_{\rm B}T_{\rm eff}$ costs energy $\sim \lambda_{\mathcal{D}}\Delta\mathcal{D}$. At sustained power P, commit rate is bounded by

$$\tau_{\rm power} = \frac{\lambda_{\mathcal{D}}\Delta\mathcal{D}}{P}$$

Coherence time ($\tau_{\rm coh}$). We model commits as threshold crossings requiring M semi-independent modules to achieve simultaneous phase alignment within tolerance ε. Let r be the mean resultant length (Kuramoto order parameter), and $\kappa(r)$ the von Mises concentration with $r = I_1(\kappa)/I_0(\kappa)$ [21].

Definition of alignment event. We define alignment as **anchored to a reference module**: all modules must fall within $\varepsilon/2$ of a designated reference phase (e.g., a pacemaker oscillator, the first module to cross threshold, or a gating signal). Formally:

$$A_\varepsilon = \{|\theta_i - \theta_1| < \varepsilon/2 \text{ for } i = 2, \ldots, M\}$$

The alignment probability has a precise geometric form. Define $p_1(\varepsilon, \kappa) = \int_{-\varepsilon/2}^{\varepsilon/2} \frac{e^{\kappa\cos\theta}}{2\pi I_0(\kappa)} d\theta$

as the single-variable window probability centered at the mode. In the small-window limit:

$$P(A_\varepsilon) \approx p_1(\varepsilon, \kappa)^{M-1} \tag{2}$$

The exponent is $(M-1)$, not M. This arises from quotient geometry: the reference module θ_1 serves as anchor, leaving $M-1$ independent constraints.

First-passage time structure. Assuming independence across modules on the commit timescale, the alignment event A_ε is a rare target set on the M-torus T^M. Under fast mixing relative to the target set size, first-passage times are approximately exponential with mean $\tau \approx 1/(\text{attempt rate} \times P(A_\varepsilon))$—a form of Kac's recurrence theorem [20]. The attempt rate is set by the phase-exploration rate $\Delta\omega$. Thus:

$$\tau_{\text{coh}} = \frac{1}{\Delta\omega}\, p_1(\varepsilon, \kappa(r))^{-(M-1)} \tag{3}$$

For coupled modules (weak inter-module coupling), the effective exponent is reduced:

$$\boxed{\tau_{\text{coh}} \approx \frac{1}{\Delta\omega}\, p_1(\varepsilon, \kappa(r))^{-\alpha(M-1)}}, \qquad \alpha \in (0, 1], \tag{4}$$

where α captures *effective independence*: $\alpha = 1$ for fully independent modules; $\alpha < 1$ when inter-module coupling induces correlations that reduce the effective number of constraints.

Physical interpretation: M is the **coordination depth**—the number of semi-independent modules whose order parameters must jointly exceed a threshold for a commit to register. $\Delta\omega$ is the **phase-exploration rate**: the characteristic frequency at which phases mix. The tolerance ε is the **full phase window width** (in radians).

2.2 The speed-flexibility trade-off

The coherence time reveals opposing constraints on commit rate:

Proposition 1 (Speed-Flexibility Frontier). *At fixed power P, systems face a Pareto frontier in (M, r) space:*

- ***Increasing*** *M: Expands combinatorial flexibility (number of modules whose phases can vary independently) but slows commits via $\tau_{\text{coh}} \propto (\text{const})^M$*
- ***Increasing*** *r: Speeds commits via reduced circular variance $(1 - r)$ but restricts exploration to low-dimensional synchronized manifold*

Increasing coordination depth M expands the dimensionality of commit-registered outcomes but requires exponentially longer alignment time. Increasing coherence r speeds commits but forces trajectories onto low-dimensional attractors, reducing the set of accessible commit outcomes.

2.3 Quantum and thermodynamic limits of computation

The reviewers correctly note that the speed of biological processing must be understood within the context of fundamental physical limits on computation. We now explicitly address the quantum and thermodynamic bounds that form the basis of information transmission in living systems.

2.3.1 The Landauer bound

Landauer's principle [11] establishes that any logically irreversible operation—specifically, the erasure or overwriting of one bit of information—dissipates at least $k_{\text{B}}T \ln 2$ of energy as heat. This bound arises from the connection between logical irreversibility and thermodynamic entropy: erasing a bit reduces the system's information-theoretic entropy, which must be compensated by entropy increase in the environment [12].

The minimum energy per bit at biological temperature ($T \approx 310$ K) is:

$$E_{\text{Landauer}} = k_{\text{B}}T \ln 2 \approx 3 \times 10^{-21} \text{ J/bit}$$

This sets an absolute floor on energy dissipation per irreversible operation. However, the Landauer bound constrains energy per operation, not time per operation. A system could in principle perform operations arbitrarily fast if sufficient power were available—until quantum speed limits intervene.

2.3.2 The Margolus-Levitin bound and quantum speed limits

The Margolus-Levitin theorem [14] establishes that the minimum time to transition between orthogonal quantum states is:

$$\tau_{\mathrm{ML}} = \frac{\pi\hbar}{2\langle E\rangle}$$

where $\langle E\rangle$ is the average energy above the ground state. This bound, together with the Mandelstam-Tamm relation [13] ($\tau_{\mathrm{MT}} = \hbar/(2\Delta E)$ where ΔE is energy uncertainty), constitutes the quantum speed limit on state evolution.

These bounds originate from the energy-time uncertainty relation $\Delta E \cdot \Delta t \gtrsim \hbar/2$, which is fundamental to understanding information transmission in physical systems [15, 16, 17]. As Bormashenko [12] notes, synthesis of the Landauer bound with the Margolus-Levitin limit yields a minimal computation time scaling as $\tau_{\min} \sim h/k_{\mathrm{B}}T$—decreasing temperature reduces energy consumption per operation but simultaneously slows computation.

At biological temperatures with $\langle E\rangle \sim k_{\mathrm{B}}T \approx 4 \times 10^{-21}$ J:

$$\tau_{\mathrm{QSL}} \sim \frac{\pi \times 10^{-34}}{2 \times 4 \times 10^{-21}} \sim 10^{-13}\ \mathrm{s}$$

This is fourteen orders of magnitude faster than the millisecond timescales of neural processing.

2.3.3 Why coherence time dominates in biological systems

The central claim of this paper is that in biological neural systems, the quantum speed limit and Landauer bound are typically **non-binding constraints**. The dominant bottleneck is coherence time—the time required for distributed oscillatory assemblies to achieve sufficient phase coordination for a commit to occur.

Consider the hierarchy of timescales for cortical visual processing:

$$\tau_{\text{QSL}} \sim 10^{-13}\ \text{s} \quad \text{(quantum speed limit)} \tag{5}$$

$$\tau_{\text{Landauer}} \sim 10^{-12}\ \text{s} \quad \text{(thermal relaxation)} \tag{6}$$

$$\tau_{\text{SNR}} \sim 10^{-3}\ \text{s} \quad \text{(signal detection)} \tag{7}$$

$$\tau_{\text{coh}} \sim 10^{-2}\ \text{s} \quad \text{(coordination time)} \tag{8}$$

The coherence time exceeds quantum limits by eleven orders of magnitude. This enormous gap arises because neural computation is not limited by how fast individual elements can switch states, but by how long it takes for *distributed assemblies of semi-independent modules* to achieve coordinated phase alignment.

Mathematically, this reflects the exponential scaling of coherence time with coordination depth:

$$\tau_{\text{coh}} \propto p_1^{-\alpha(M-1)}$$

For $M = 8$ modules with $p_1 \approx 0.7$ and $\alpha = 0.7$, the exponential factor is $(0.7)^{-4.9} \approx 10$. Even modest coordination requirements ($M \sim 5$–15) produce millisecond-scale coherence times from microsecond-scale phase dynamics.

This analysis clarifies the relationship between our framework and fundamental physics: quantum and thermodynamic limits set absolute bounds on computation, but biological systems operate in a regime where multi-module coordination—not quantum switching speed or Landauer dissipation—determines processing rate.

2.4 Parameter estimation protocols

Coordination depth (M). M is the count of semi-independent modules whose order parameters must jointly exceed a threshold to register a commit. **Operational estimation:**

1. Cluster channels/units by band-limited coherence or Granger causality into modules.
2. Identify the minimal set of modules whose joint activity predicts behavioral commits.
3. Typical values: occipital visual tasks yield $M \sim 6$–10; cross-modal binding $M \sim 10$–15.

Kuramoto coherence (r). Extract instantaneous phase $\theta_j(t)$ from bandpass-filtered LFP/EEG via Hilbert transform. Compute

$$r(t) = \left| \frac{1}{N} \sum_{j=1}^{N} e^{i\theta_j(t)} \right|$$

Typical values: spontaneous cortex ~ 0.2–0.4; attention ~ 0.5–0.7; circadian networks ~ 0.7–0.9 [2].

Phase alignment window (ε). From spike train or LFP cross-correlograms, measure half-width at half-maximum (HWHM, units: seconds). The **full** phase window is $\varepsilon = 2\,\omega_0\,\mathrm{HWHM}$ (radians).

Phase-exploration rate ($\Delta\omega$). Estimate from instantaneous frequency spread or phase diffusion. Typical values: $\Delta\omega \sim 2\pi \times (5\text{–}20)$ rad/s for cortical gamma/alpha bands.

3 Results

3.1 Visual perceptual binding windows

Human visual perception exhibits temporal integration windows of 30–50 ms (flicker fusion at 20–30 Hz). We apply Eq. 1 with neural parameters.

Take $M = 8$ occipital modules, $r = 0.7$ (attention [2]), full tolerance $\varepsilon = \pi$ rad, and phase-exploration rate $\Delta\omega = 2\pi \times 20$ rad/s.

First compute p_1: at $r = 0.7$, $\kappa \approx 2.0$, giving $p_1(\pi, 2.0) \approx 0.62$ (numerical integration). Using Eq. 4 with $\alpha = 0.7$ (modular cortical topology):

$$\tau_{\mathrm{coh}} \approx \frac{1}{125.66}(0.62)^{-0.7\times 7} = \frac{1}{125.66}(0.62)^{-4.9} \approx 83 \text{ ms}.$$

With higher coherence $r = 0.8$ ($\kappa \approx 3.0$, $p_1(\pi, 3.0) \approx 0.76$):

$$\tau_{\mathrm{coh}} \approx \frac{1}{125.66}(0.76)^{-4.9} \approx 30 \text{ ms}.$$

Computing all terms in Eq. 1:

$$\tau_{\mathrm{QSL}} \sim 10^{-13}\,\text{s} \quad \text{(negligible—quantum limit non-binding)} \tag{9}$$

$$\tau_{\mathrm{SNR}} \sim 5\text{–}10 \text{ ms} \quad \text{(task-dependent)} \tag{10}$$

$$\tau_{\mathrm{power}} \ll 1 \text{ ms} \quad \text{(per commit)} \tag{11}$$

$$\tau_{\mathrm{coh}} \approx 30\text{–}83 \text{ ms} \quad (r = 0.7\text{–}0.8,\ M = 8) \tag{12}$$

The effective commit time is $\tau_{\mathrm{eff}} = \max[\mathrm{QSL}, \mathrm{SNR}, \mathrm{power}, \mathrm{coh}] \approx 30$–$83$ ms, matching human binding windows (30–50 ms). **Coherence time dominates**; quantum and Landauer bounds are not the limiting factors.

3.2 Tachypsychia: dual-loop dissociation

During acute stress or falls, subjects report subjective time slowing while objective reaction times remain unchanged [22]. This dissociation suggests separate commit pathways:

Perceptual loop (cortical): High-M ($\sim$ 10–15 modules) coherent field dynamics. Commits sparse (5–20 Hz). Arousal increases coherence r and thus information rate.

Motor loop (cerebellar/basal ganglia): Low-M ($\sim$ 3–5 modules) primitives. Commits faster (50–150 ms). Arousal modulates decision threshold, preserving reaction time.

This dual-loop architecture explains the dissociation: perceptual commits (high M) proceed independently of motor commits (low M).

3.3 Alpha oscillations and temporal acuity

Alpha oscillations (8–12 Hz) in visual cortex correlate with temporal acuity: individuals with faster alpha perceive time faster [23, 24]. We interpret alpha as reflecting commit frequency: $r_{\text{commit}} \sim f_\alpha$.

Prediction: Manipulating alpha via transcranial alternating current stimulation (tACS) should proportionally shift perceptual windows.

3.4 Metabolic scaling across species

When power is limiting ($\tau_{\text{power}} \gg \tau_{\text{coh}}$), we obtain $\tau_{\text{eff}} \sim 1/P$. Critical flicker fusion frequency should scale with mass-specific metabolic rate.

Healy et al. [25] measured critical flicker fusion frequency across diverse taxa: flies ~ 240 Hz, humans ~ 60 Hz, leatherback turtles ~ 15 Hz. Log-log regression shows $R^2 \approx 0.6$ across three orders of magnitude in body mass, validating the power-limited regime for small, high-metabolism animals.

4 Discussion

4.1 Biological examples of the speed-flexibility trade-off

Mind-wandering and creative insight (high M, low r). Default-mode network activity exhibits low inter-regional coherence ($r \sim 0.3$) and high coordination depth ($M \sim 12$–20). Commits are rare (~ 0.1–1 Hz), producing subjective "slow thinking" but enabling novel associations.

Focused attention (moderate M, moderate r). Attention increases coherence (r : $0.3 \to 0.6$) and narrows coordination to task-relevant modules (M : $15 \to 8$) [2]. This produces faster commits (10–30 Hz) in a restricted subspace.

Automaticity and skill learning (low M, high r). Overlearned motor sequences compress to low-dimensional primitives ($M \sim 3$–5) with tight coherence ($r \sim 0.8$). Commits are rapid (50–150 ms) but inflexible.

4.2 Temporal embedding as a pre-commit representation mechanism

The speed-flexibility trade-off explains *when* oscillatory assemblies can commit, but not *how* high-dimensional structure is preserved during the long pre-commit interval. We propose that biological systems implement **Takens delay embedding** [26].

Takens' theorem. For a deterministic system with state dimension d, time-delayed copies of a scalar observable—$x(t), x(t-\tau), x(t-2\tau), \ldots$—can reconstruct the complete state-space topology. Crucially, $2d+1$ delay coordinates generically suffice.

Biological implementation: cerebellar parallel fibers. Parallel fibers—unmyelinated granule cell axons—conduct at 0.2–0.5 m/s with systematically varying path lengths, creating conduction delays of 5–15 ms [27, 28, 29]. These delays match the optimal embedding parameter $\tau^* = 1/(4f)$ for motor dynamics at 15–50 Hz.

We hypothesize that the cerebellum provides a concrete biological example of delay-embedded preservation of high-dimensional dynamics during pre-commit intervals, enabling rapid low-dimensional readout without requiring simultaneous global coordination.

Robustness to noise. We use Takens embedding as a *design principle* rather than a literal guarantee: delay-line architectures improve state inference under partial observability, even when exact reconstruction is impossible.

Connection to coherence time. Coherence time determines the maximum window over which temporal embeddings remain usable. If coherence decays faster than the embedding window, reconstruction fails and high-dimensional structure is lost before a commit can occur.

4.3 Generalization beyond neural systems

While we emphasize neural applications given their empirical accessibility, the coherence time framework applies to any biological system whose function requires coordinated phase relations and whose outputs are sparse commitment events.

Scope statement. Neural systems are the best-instrumented instance of a broader class: living, self-maintaining oscillator assemblies that perform irreversible state registration. We do not treat collectives (e.g., societies) as intelligent systems in this sense; they are coordination networks of intelligent organisms and introduce additional channels not modeled here.

Chromosomal coordination. Chromosomal systems provide a non-neural example of the same architecture. High-dimensional continuous dynamics (polymer conformations, electrostatic interactions, topological constraints, phase separation) produce function via sparse low-dimensional commits (origin firing, checkpoint decisions, segregation outcomes).

Anaphase onset is naturally modeled as a commit event: a global irreversible transition triggered only after multiple semi-independent units satisfy a coherence-like constraint. The waiting time should scale steeply with effective coordination depth, consistent with first-passage into a small target set. The relevant "order parameter" is any coordination metric determining whether a global event can proceed—spatial colocalization, topological state, mechanical alignment—not necessarily Kuramoto r.

Circadian networks. Circadian gene networks exhibit $M \sim$ 3–5 core feedback loops, $r \sim 0.8$ (tight coupling), and $\varepsilon \sim 0.5$ rad (loose phase tolerance). These parameters yield coordination times much shorter than the 24-hour period, implying coordination is *not* the bottleneck. The 24-hour period arises from molecular lags. The framework correctly identifies the dominant regime: circadian clocks are biochemical-delay-limited, not coordination-limited.

4.4 Testable predictions

Alpha entrainment. tACS at alpha frequency should shift perceptual binding windows linearly with frequency change. 20% frequency increase predicts 20% threshold reduction.

Arousal effects on dual-loop dissociation. During simultaneous simple reaction time + temporal-order judgment under acute stress:

- Temporal-order discrimination threshold changes
- Simple reaction time unchanged or slightly faster
- Statistical independence between measures

Coherence measures and consciousness. If phenomenal experience reflects pre-commit coherent dynamics, conscious states should correlate more strongly with phase-locking index and theta-gamma coupling than with spike count or mean firing rate.

4.5 Relationship to existing frameworks

Quantum-like cognition. Recent work has explored connections between oscillatory neural networks and quantum-like cognitive phenomena [18]. Our framework provides a classical high-dimensional explanation for why such quantum-like signatures emerge: high-dimensional dynamics with limited measurement bandwidth produce phenomenology parallel to quantum systems—measurement-like collapse at commits, superposition-like coexistence of states pre-commit, uncertainty-like timing inaccessibility.

Integrated Information Theory [30]: We provide thermodynamic/temporal foundations. Integration costs energy and takes time (exponentially with coordination depth).

Free Energy Principle [31]: We add temporal structure. Prediction updates occur at commit events with rate limited by Eq. 1.

5 Conclusion

We have established that coherence time—the time required for distributed oscillatory assemblies to achieve sufficient phase alignment for irreversible state registration—bounds the rate of neural processes associated with cognition. The bound

$$\tau_{\text{eff}} = \max\left[\tau_{\text{QSL}},\, \tau_{\text{SNR}},\, \tau_{\text{coh}},\, \tau_{\text{power}}\right]$$

with coherence time

$$\tau_{\text{coh}} = \frac{1}{\Delta\omega}\, p_1(\varepsilon, \kappa(r))^{-\alpha(M-1)}$$

unifies quantum, noise, and coordination limits to explain temporal resolution in biological processing systems.

We have situated this framework explicitly within the hierarchy of physical limits on computation: the Landauer bound constrains energy per irreversible operation; the Margolus-Levitin and Mandelstam-Tamm bounds constrain the speed of quantum state evolution; the energy-time uncertainty relation underlies both. In biological neural systems, these quantum and thermodynamic bounds are typically non-binding. The dominant constraint is the time required for multi-module coordination—a classical phenomenon arising from rare-event statistics on product manifolds.

Key findings:

1. **Speed-flexibility trade-off**: Increasing M exponentially slows commits but expands combinatorial flexibility. Increasing r speeds commits but restricts dynamics to low-dimensional manifolds.

2. **Quantum limits non-binding**: Neural coherence times exceed quantum speed limits by eleven orders of magnitude. Coordination, not quantum switching, limits processing rate.

3. **Generality**: While neural systems provide the empirical testbed, the framework applies to any biological oscillator assembly with coordinated phase relations and sparse commits.

4. **Quantitative predictions**: Visual binding windows (30–50 ms), metabolic scaling ($R^2 = 0.6$), alpha entrainment linearity, dual-task dissociations.

The central thesis: biological systems trade speed for flexibility by moving on the coherence-dimension frontier, operating in a regime where quantum and thermodynamic limits permit far faster processing than coordination constraints allow.

Data accessibility

This is a theoretical framework paper. Kuramoto simulation code and parameter estimation workflows are available at `https://github.com/todd866/coherence-time`.

Competing interests

The author declares no competing interests.

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